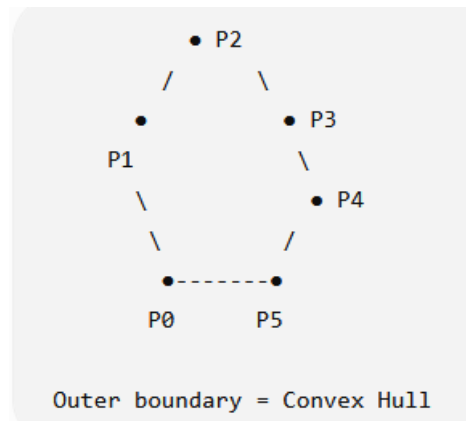


Chapter-10

1. Convex Hull

A Convex Hull is the smallest convex polygon that completely encloses all the control points of a curve. A shape is convex if a straight line drawn between any two points inside the shape always stays completely inside the shape. The convex hull helps designers understand the overall shape of the curve and is commonly used in spline curve modeling.



Characteristics

- It is the smallest convex polygon that completely encloses all the given points.
- The vertices of the convex hull are always selected from the original set of points.
- All control points lie either inside the convex hull or on its boundary.
- Many spline curves, such as Bézier curves, remain completely inside the convex hull.
- It helps prevent unwanted curve oscillations and makes the curve shape more predictable.
- It is widely used in computer graphics for collision detection, clipping, pattern recognition, and shape analysis.

2. Parametric Continuity Conditions

Parametric continuity ensures that two curve segments join smoothly by matching their mathematical derivatives at the connecting point. The higher the order of continuity, the smoother the curve becomes. This continuity is widely used in computer graphics, CAD, and animation.

Types

C⁰ Continuity (Zero-order)

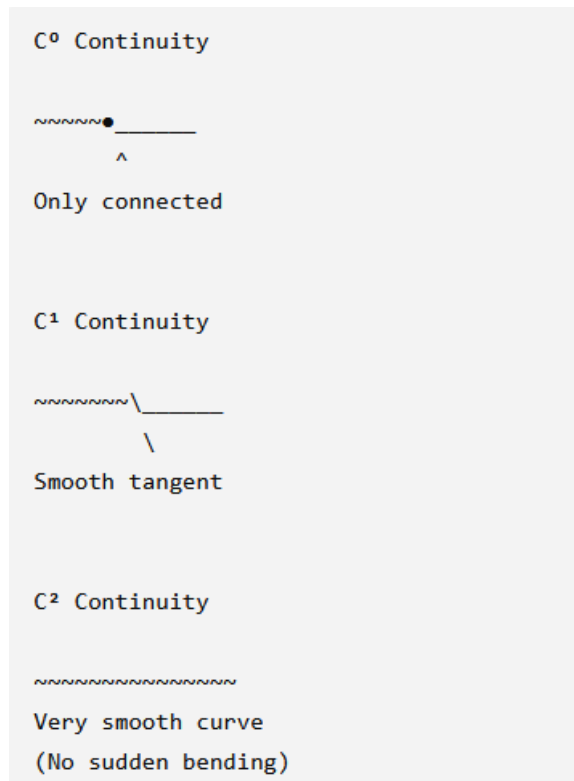
- Curves meet at the same point.
- Only position is continuous.
- No guarantee of smoothness.

C¹ Continuity (First-order)

- Position and tangent are continuous.
- First derivatives are equal.
- Produces a smooth direction change.

C² Continuity (Second-order)

- Position, tangent, and curvature are continuous.
- First and second derivatives are equal.
- Produces a very smooth curve.



3. Geometric Continuity Conditions

Geometric Continuity (G-continuity) is a method of joining two curve segments so that they appear smooth. Unlike parametric continuity, the tangent vectors do not need to be equal—they only need to have the same direction (be proportional). This produces a smooth-looking curve and is widely used in computer graphics and CAD.

Types of Geometric Continuity

G⁰ Continuity (Zero-Order)

- Two curve segments meet at the same point.
- Only the position is continuous.
- No smooth transition.

G¹ Continuity (First-Order)

- Curves meet at the same point.
- Tangent directions are the same.
- The curve looks smooth, but the speed of change may be different.

G² Continuity (Second-Order)

- Position, tangent direction, and curvature are continuous.
- Produces a very smooth curve.
- Commonly used in high-quality design and animation.

Characteristics

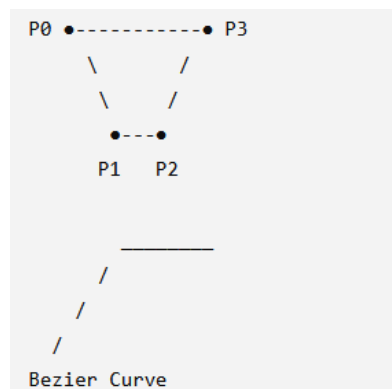
- Creates a smooth visual connection between curves.
- Uses proportional tangent vectors instead of equal tangent vectors.
- More flexible than parametric continuity.
- Produces natural-looking curves.

4. Bezier Curves and Its Properties

A **Bezier Curve** is a smooth parametric curve created using a set of **control points**. The first and last control points determine the starting and ending points of the curve, while the other control points control its shape. Bezier curves are widely used in computer graphics because they are easy to design and produce smooth curves.

Properties of Bezier Curves

- Passes through the first and last control points.
- Usually does not pass through the intermediate control points.
- Lies completely inside the convex hull of its control points.
- The degree of the curve is one less than the number of control points.
- Moving a control point changes the shape of the curve.
- Easy to scale, rotate, and transform.



Explanation of Diagram

- P0 = Starting control point
- P3 = Ending control point
- P1 and P2 = Control the shape of the curve

5. Spline Curves

A Spline Curve is a smooth curve created by joining several polynomial curve segments together. It is defined using a set of control points, which determine the overall shape of the curve. Splines are widely used in computer graphics because they produce smooth curves that are easy to modify. They are commonly used in CAD, animation, and surface modeling.

Types of Spline Curves

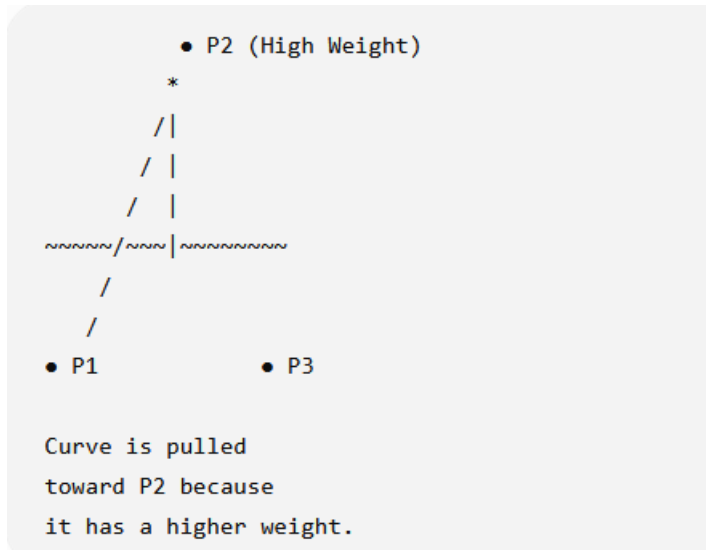
- **Interpolation Splines** – Pass through all control points.
- **Approximation Splines** – Follow the general shape of the control points but do not necessarily pass through them.

Characteristics

- Produces smooth and continuous curves.
- Defined using control points.
- Easy to modify by moving control points.
- Can represent complex shapes accurately.
- Supports local or global curve control depending on the spline type.

6. Rational Splines

A Rational Spline is a type of spline curve that uses control points and weights to define its shape. Each control point has a weight, and a control point with a higher weight pulls the curve closer to itself. Rational splines are more flexible than ordinary splines and can represent circles, ellipses, and other curved shapes accurately.



Properties

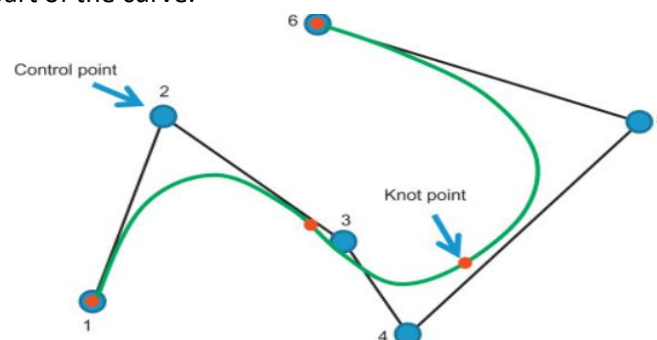
- Uses weighted control points.
- Higher weight pulls the curve closer to the control point.
- Can represent circles and ellipses exactly.
- Provides greater flexibility than ordinary splines.
- Produces smooth curves.

Advantages

- More accurate curve representation.
- Better shape control.
- Suitable for engineering designs.
- Can model complex objects.

7. B-Spline Curves

A B-Spline (Basis Spline) is a smooth curve created using a set of control points. Unlike a Bezier curve, a B-Spline usually does not pass through all the control points. Instead, the control points influence the shape of the curve. B-Spline curves provide local control, meaning that moving one control point changes only a small part of the curve.



Properties

- Defined using a set of control points.
- Usually does not pass through all control points.
- Produces smooth and continuous curves.
- Lies inside the convex hull of its control points.
- Provides local control over the curve shape.
- Moving one control point affects only a portion of the curve.
- More flexible than Bezier curves.
- Easy to modify without changing the entire curve.

8. Beta-Splines

A Beta-Spline is an improved version of the B-Spline that introduces beta parameters to control the tension and shape of the curve. It combines the smoothness of B-Splines with greater flexibility in curve design. It uses two parameters, bias (β_1) and tension (β_2), to adjust the curve without changing the control points.

Parameters

- **Bias (β_1):** Controls the direction or skewness of the curve.
- **Tension (β_2):** Controls how tight or loose the curve is.

Properties

- It is a smooth and flexible spline curve.
- It uses bias and tension parameters to control the curve shape.
- The curve does not usually pass through the control points.
- Changing the parameters changes the curve shape without moving the control points.
- It provides better local control than many other spline curves.
- It is suitable for designing smooth and natural-looking curves.

9. Ray-Casting Method

The Ray-Casting Method is an algorithm used to determine whether a point lies inside or outside a polygon. It works by drawing a ray (a straight line) from the point in any direction, usually to the right, and counting how many times the ray intersects the edges of the polygon.

- If the number of intersections is **odd**, the point is **inside** the polygon.
- If the number of intersections is **even**, the point is **outside** the polygon.

Steps

1. Draw a ray from the given point to infinity (usually horizontally to the right).
2. Count the number of times the ray intersects the polygon's edges.
3. If the count is **odd**, the point is **inside** the polygon.
4. If the count is **even**, the point is **outside** the polygon.

Properties

- It is a simple and efficient point-in-polygon test.
- It works for both convex and non-convex polygons.
- The result depends on whether the number of edge intersections is odd or even.
- It is easy to implement in computer graphics applications.
- Special care is needed when the ray passes through a polygon vertex or lies along an edge.

10. Octrees

An Octree is a hierarchical tree-based data structure used to divide three-dimensional (3D) space into smaller and more manageable regions. It starts with one large cube representing the entire space. If a region contains too many objects or needs more detail, it is recursively divided into eight equal-sized smaller cubes, called children. This process continues until the required level of detail is achieved. Octrees provide an efficient way to organize, store, search, and process 3D objects, making them widely used in computer graphics, game development, collision detection, ray tracing, and spatial indexing.

Properties

- It divides 3D space into eight equal parts at each level.
- Each internal node has up to eight children.
- It represents space in a hierarchical tree structure.
- Empty regions do not need to be stored, saving memory.
- It provides efficient searching and spatial organization.

11. BSP Trees (Binary Space Partition Trees)

A BSP Tree (Binary Space Partition Tree) is a hierarchical tree-based data structure used to divide 2D or 3D space into smaller regions using partitioning planes (or lines in 2D). Each partition splits the space into two parts, creating a binary tree. This process continues recursively until the space is divided into manageable regions. BSP trees efficiently organize geometric objects, making operations such as rendering, visibility determination, collision detection, and spatial searching faster.

Properties

- Divides space into two regions at each partition.
- Uses partitioning planes in 3D and lines in 2D.
- Forms a hierarchical binary tree structure.
- Organizes objects based on their spatial location.
- Improves searching, visibility testing, and rendering efficiency.
- Supports recursive subdivision of space.

12. Zero-Order Continuity (C^0 / G^0)

Definition

Zero-order continuity means that two curve segments **meet at the same point**. The position is continuous, but the direction and smoothness may change suddenly. It is the simplest type of continuity.

Characteristics

- Curves are connected.
- Position is continuous.
- Tangent may change abruptly.
- No smooth transition.

13. First-Order Continuity (C^1 / G^1)

Definition

First-order continuity means that two curve segments meet and have the same tangent direction at the joining point. This creates a smooth transition between the curves, although the curvature may still change.

Characteristics

- Position is continuous.
- Tangent direction is continuous.
- Smooth change in direction.
- Better appearance than C^0 .

14. Second-Order Continuity (C^2 / G^2)

Definition

Second-order continuity means that two curve segments have continuous position, tangent, and curvature. Both the first and second derivatives are continuous, producing a very smooth curve. It is commonly used where high precision and smooth motion are required.

Characteristics

- Position is continuous.
- Tangent is continuous.
- Curvature is continuous.
- Produces very smooth curves.
- Eliminates sudden changes in acceleration.